Riki Shimizu Ph.D. Candidate

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Interests

I am an **audio AI and neural engineering researcher**. I develop real-time wearable speech enhancement systems, with the goal of making them controllable through brain signals. My research also spans audio model interpretability and multimodal LLMs, including post-training and evaluation. *The icons mark these research themes in the publications below.*

Schools

Columbia University — New York, NY, US
Ph.D. Candidate in Electrical Engineering, after joint M.S. — *2024.08 – 2029 (expected)*

- Advisor: Prof. Nima Mesgarani
- Research: Speech AI, Brain-Computer Interface, and Multimodal LLMs

Duke University — Durham, NC, US
B.S.E. in Biomedical Engineering — *2020.08 – 2024.05*
B.S.E. in Electrical & Computer Engineering
B.S. in Mathematics with Distinction

- Advisors: Prof. Pratik Chhatbar, Prof. Leslie Collins, and Prof. Hau-Tieng Wu
- Research: Brain-Computer Interface and Signal Processing
- Minors: Computer Science and Philosophy

Interns

Meltin MMI Co., Ltd. — Tokyo, Japan
Software Engineering Intern — *2022.05 – 2022.08*
Developed Python models of nerve regeneration with sieve electrodes using diffusion-reaction and stochastic propagation, reducing the electrode prototype testing cycle from months to hours.

M3, Inc. — Tokyo, Japan
Market Analysis & Consulting Intern — *2020.05 – 2020.07*
Conducted market research on AI diagnostic imaging using the FDA device database and supported PMDA regulatory submissions.

School Services

Teaching Assistant, Columbia University — 2026.01 – 2026.05
Advanced Big Data and AI (EECS E6895)
Instructor: Prof. Ching-Yung Lin

Papers

J denotes journal articles; C denotes conference papers/presentations.
** denotes equal contributions.*

Speech Enhancement System
Shimizu, R., et al. (2026). *Under review at ACM CHI*. Project to be disclosed.

[C10] Cocktail-Talker: Multi-Speaker Dialog Modeling in Noisy Social Environments with Turn Action GRPO
Jiang, X., **Shimizu, R.**, Dindar, S. S., Wu, J., Xu, Z., and Mesgarani, N. (2026). *SLT 2026*. Accepted. [pdf]

Sympatheia: Emotionally Adaptive Voice Assistant with Continuous Affect Conditioning
Dindar, S. S., **Shimizu, R.**, Jiang, X., and Mesgarani, N. (2026). *in review*, [pdf]

[J3] Interpretable Embeddings of Speech Enhance and Explain Brain Encoding Performance of Audio Models
Shimizu, R., Antonello, R. J., Singh, C., and Mesgarani, N. (2026). *PLOS Computational Biology*. [pdf] [code]

[C9] MeanFlow-TSE: One-Step Generative Target Speaker Extraction with Mean Flow
Shimizu, R., Jiang, X., and Mesgarani, N. (2026). *2026 INTERSPEECH*, [pdf] [code]

[C8] AVMeme Exam: A Multimodal Multilingual Multicultural Benchmark for LLMs' Contextual and Cultural Knowledge and Thinking
Jiang, X., Wang, Q., Wu, J., He, X., Xu, Z., Ma, Y., Piao, M., Yang, K., Zheng, X., **Shimizu, R.**, Chen, Y., Firoozi, A., Mischler, G., Dindar, S. S., Antonello, R., He, L., Hsieh, T.-A., Fan, X., Wu, Y., Ma, Y., Amballa, C., Chen, W., Hai, J., Li, R., Choudhari, V., Han, C., Li, Y. A., Flinker, A., Elhilali, M., Benetos, E., Hasegawa-Johnson, M. A., Roy Choudhury, R., and Mesgarani, N. (2026). *2026 Conference on Language Modeling (COLM)*, [pdf] [data]

[C7] Interpretable Embeddings of Speech Explain and Enhance the Brain Encoding Performance of Audio Models
Shimizu, R., Antonello, R. J., Singh, C., and Mesgarani, N. (2026). *Association for Research in Otolaryngology (ARO)*. **Oral presentation.**

[J2] C3-C4 Transcranial Direct Current Stimulation Montage Stimulates Lower Limb Region Better than C1-C2 Montage
Huang, Z.*, **Shimizu, R.***, Sinha, S. R., Southwell, D., Charalambous, C. C., Peterchev, A. V., Feng, W., Chhatbar, P. Y., and Kim, T. (2025). *Clinical Neurophysiology*, [pdf]

[C6] XCOMPS: A Multilingual Benchmark of Conceptual Minimal Pairs
He, L., Nie, E., Dindar, S. S., Firoozi, A., Florea, A., Nguyen, V., Puffay, C., **Shimizu, R.**, Ye, H., Brennan, J., Schmid, H., Schütze, H., and Mesgarani, N. (2025). *7th Workshop on Research in Computational Linguistic Typology and Multilingual NLP (SIGTYP)*, [pdf]

[C5] An Assessment of the Impact of Feature Preprocessing on Deep Learning Models for P300 Classification
Shimizu, R., Collins, L., and Mainsah, B. (2025). *11th International BCI Meeting*, [pdf]

[J1] Unveil Sleep Spindles with Concentration of Frequency and Time (ConceFT)
Shimizu, R., and Wu, H.-T. (2024). *Physiological Measurement*, [pdf] [code]

[C4] Individualized Electric Field Modeling for Transcranial Electrical Stimulation in Stroke Patients
Huang, Z., **Shimizu, R.**, Feld, J., Sokhadze, E., Chhatbar, P. Y., and Feng, W. (2023). *International Society for Neuroregulation & Research (ISNR)*

[C3] Personalized Electric Field Modeling for Transcranial Electrical Stimulation on Stroke Patients: A Comparison of Open-Source Pipelines, With Versus Without Manual Stroke Delineation
Huang, Z., **Shimizu, R.**, Feld, J., Sokhadze, E., Chhatbar, P. Y., and Feng, W. (2023). *International Brain Stimulation Conference*

[C2] Transcranial Magnetic Stimulation Responses Show Differential Recovery Pattern Between Upper and Lower Limb in Stroke Patients Over Subacute Phase
Kim, T., **Shimizu, R.**, Malkan, P., Park, C., Heffron, L., Penner, A., Feld, J., Feng, W., and Chhatbar, P. Y. (2021). *International Brain Stimulation Conference*

[C1] Interhemispheric Inhibition Findings Using Ipsilateral Silent Period in Two Stroke Patients During Acute and Subacute Phases After Stroke
Chhatbar, P. Y., Kim, T., Malkan, P., **Shimizu, R.**, Park, C., Heffron, L., Penner, A., Feld, J., and Feng, W. (2021). *International Brain Stimulation Conference*

Skills

Languages: Japanese (native), English (bilingual proficiency)

Programming: Python, PyTorch, HuggingFace, SpeechBrain, MLX, ONNX, MATLAB, C

Others: Linux, Slurm, multi-GPU training, git, LaTeX

Honors

- Blavatnik Presidential Fellowship — 2024–present
  The most prestigious PhD fellowship given by Columbia Engineering.
- PRUV Mathematics Fellowship — 2023–2024
  Duke University; competitive undergraduate research fellowship ($3,500).

- Data+ Research Grant 2022–2023
  Rhodes Information Initiative at Duke; summer research grant for data science projects ($5,000).
- Tau Beta Pi Engineering Honor Society 2022–present
  Inducted as a member of the top 1/8 of the engineering class at Duke University.
- Yanai Tadashi Foundation Scholarship 2020–2024
  One of 25 recipients selected nationally in Japan; funded by Tadashi Yanai, founder of UNIQLO, covering college tuition and board ($380,000).
- Japan Representative, International Brain Bee 2019
  Represented Japan at the highest-level international neuroscience academic competition for high school students; placed 18th out of 25,000+ competitors worldwide.

Community Service

Common Denominator and Reading Team Math 2024.08 – present
Teach weekly math lessons to underserved elementary school students in Harlem.

Saturday Science Volunteer 2024.08 – present
Teach basic neuroscience concepts to children in Harlem.